Let

$$x = \begin{pmatrix} -2 \\ 6 \\ 6 \end{pmatrix}, \quad y = \begin{pmatrix} 0 \\ -12 \\ \alpha \end{pmatrix}, \quad z = \begin{pmatrix} \beta \\ \gamma \\ 6 \end{pmatrix}$$

Find the values of α, β, γ to be x, y, z pairwise orthogonal.

$\alpha =$ 12

$\beta =$ 36

$\gamma =$ 6

$\langle x, y\rangle = 0$
$\langle x, z\rangle = 0$
$\langle y, z\rangle = 0$

$$\begin{cases} -72 + 6\alpha = 0 \rightarrow \alpha = 12 \\ -2\beta + 6\gamma + 36 = 0 \rightarrow -2\beta + 36 + 36 = 0 \rightarrow 2\beta = 72 \rightarrow \beta = 36 \\ -12\gamma + 6\alpha = 0 \rightarrow -12\gamma + 72 = 0 \rightarrow \gamma = 6 \end{cases}$$

$\lambda = 1$ is an eigenvalue of

$$A = \begin{pmatrix} -3 & 4 & 4 \\ -1 & 1 & -1 \\ -1 & 2 & 4 \end{pmatrix}.$$

Complete the vector x to get an eigenvector corresponding to λ. (Round the result to 2 decimal places.)

$x_1 = 1$

$x_2 =$ 2

$x_3 =$ -1

$\lambda = 1$

$$\begin{pmatrix} -3-1 & 4 & 4 \\ -1 & 1-1 & -1 \\ -1 & 2 & 4-1 \end{pmatrix} \begin{pmatrix} 1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\begin{cases} -4 + 4x_2 + 4x_3 = 0 \\ -1 - x_3 = 0 \\ -1 + 2x_2 + 3x_3 = 0 \end{cases}$$

$x_3 = -1$

$-1 + 2x_2 - 3 = 0$

$2x_2 = 4$

$x_2 = 2$

If

$$x = \begin{pmatrix} -3 \\ -4 \\ 0 \end{pmatrix}, \quad \text{and} \quad y = \begin{pmatrix} -8 \\ 1 \\ -7 \end{pmatrix}$$

then the dot product of x and y is

Answer: 20

$\langle x, y\rangle = 0$

$24 - 4 = 20$

Let

$$A = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -4 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 1 & -3 \\ 0 & 5 & -2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Give the eigenvalues of A in ascending order.

-4

-2

-1

Give the eigenvalues of B in ascending order.

1

4

5

$\det(A) = (-2-\lambda)(-1-\lambda)(-4-\lambda) = 0$

$-2-\lambda = 0 \quad -1-\lambda = 0 \quad -4-\lambda = 0$

$\lambda = -2 \quad \lambda = -1 \quad \lambda = -4$

$\det(B) = (4-\lambda)(5-\lambda)(1-\lambda) = 0$

Calculate the Euclidean norm of

$$x = \begin{pmatrix} 1 \\ 5 \\ -9 \end{pmatrix},$$

rounded to 2 decimal places.

Answer: 10.34

$\sqrt{1+25+81} = \sqrt{107} = 10.34\cdots$

Find the value of α, if the vectors x and y are orthogonal.

$$x = \begin{pmatrix} 0 \\ -5 \end{pmatrix}, \quad y = \begin{pmatrix} -20 \\ \alpha \end{pmatrix}$$

Answer: 0

$\langle x, y \rangle = 0$

$0 - 5\alpha = 0$

$-5\alpha = 0$

$\alpha = 0$

List the eigenvalues of

$$A = \begin{pmatrix} 21 & -6 \\ 60 & -17 \end{pmatrix}$$

in ascending order.

$\lambda_1 =$ 1

$\lambda_2 =$ 3

Complete the coordinates of x to obtain an eigenvector of A associated with λ_1. (Give the result in decimal, rounded to 2 decimal places.)

$x_1 = 1$

$x_2 =$ 3.33

$\begin{vmatrix} 21-\lambda & -6 \\ 60 & -17-\lambda \end{vmatrix} = 0$

$(21-\lambda)(-17-\lambda) - (-360) = 0$

$-357 - 21\lambda + 17\lambda + \lambda^2 + 360 = 0$

$\lambda^2 - 4\lambda + 3 = 0$

$\begin{matrix} 1 \\ 1 \end{matrix} \times \begin{matrix} -3 \\ -1 \end{matrix}$

$(\lambda - 3)(\lambda - 1) = 0$

$\lambda = 1, 3$

$\lambda_1 = 1$

$\begin{pmatrix} 20 & -6 \\ 60 & -18 \end{pmatrix} \begin{pmatrix} 1 \\ x_2 \end{pmatrix} = 0$

$\begin{cases} 20 - 6x_2 = 0 \\ 60 - 18x_2 = 0 \end{cases} \leftarrow$

$6x_2 = 20$

$x_2 = \frac{10}{3}$

$= 3.33\ldots$

List the eigenvalues of

$$A = \begin{pmatrix} 19 & -8 \\ 40 & -17 \end{pmatrix}$$

in ascending order.

-1

3

$\begin{vmatrix} 19-\lambda & -8 \\ 40 & -17-\lambda \end{vmatrix} = 0$

$(19-\lambda)(-17-\lambda) - (-320) = 0$

$-323 - 19\lambda + 17\lambda + \lambda^2 + 320 = 0$

$\lambda^2 - 2\lambda - 3 = 0$

$\begin{matrix} 1 \\ 1 \end{matrix} \times \begin{matrix} -3 \\ 1 \end{matrix}$

$(\lambda - 3)(\lambda + 1) = 0$

$\lambda = -1, 3$

Find the value of α, if the vectors x and y are orthogonal.

$$x = \begin{pmatrix} 9 \\ -2 \end{pmatrix}, \quad y = \begin{pmatrix} 8 \\ \alpha \end{pmatrix}$$

Answer: 36

$\langle x, y \rangle = 0$

$72 - 2a = 0$

$2a = 72$

$a = 36$

Let

$$x = \begin{pmatrix} -4 \\ -16 \\ -8 \end{pmatrix}, \quad y = \begin{pmatrix} 0 \\ -24 \\ \alpha \end{pmatrix}, \quad z = \begin{pmatrix} \beta \\ \gamma \\ 8 \end{pmatrix}$$

Find the values of α, β, γ to be x, y, z pairwise orthogonal.

$\alpha =$ 48

$\beta =$ −80

$\gamma =$ 16

$\langle x, y \rangle = 0$: $384 - 8a = 0 \rightarrow 8a = 384 \rightarrow a = 48$

$\langle x, z \rangle = 0$: $-4\beta - 16\gamma - 64 = 0 \rightarrow -4\beta = 320 \rightarrow \beta = -80$

$\langle y, z \rangle = 0$: $-24\gamma + 8a = 0 \rightarrow -24\gamma + 384 = 0 \rightarrow \gamma = 16$

$\lambda = -1$ is an eigenvalue of

$$A = \begin{pmatrix} -3 & 4 & 4 \\ -1 & 1 & -1 \\ -1 & 2 & 4 \end{pmatrix}.$$

Complete the vector x to get an eigenvector corresponding to λ. (Round the result to 2 decimal places.)

$x_1 =$ ~~−1~~ **2**

$x_2 = 1$

$x_3 =$ ~~−1~~ **0**

$\lambda = -1$

$$\begin{pmatrix} -3+1 & 4 & 4 \\ -1 & 1+1 & -1 \\ -1 & 2 & 4+1 \end{pmatrix} \begin{pmatrix} x_1 \\ 1 \\ x_3 \end{pmatrix} = 0$$

$\cancel{-4x_1 + 4 + 4x_3 = 0}$

$-x_1 - 1 = 0 \rightarrow x_1 = -1$

$-x_1 + 2 + 3x_3 = 0 \rightarrow -(-1) + 2 + 3x_3 = 0$

$3x_3 = -3$

$x_3 = -1$

$$\begin{pmatrix} -2 & 4 & 4 \\ -1 & 2 & -1 \\ -1 & 2 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ 1 \\ x_3 \end{pmatrix} = 0$$

$-2x_1 + 4 + 4x_3 = 0$

$-x_1 + 2 - x_3 = 0$

$-x_1 + 2 + 5x_3 = 0$

$-x_1 + 2 - x_3 = 0$

$-)\ -x_1 + 2 + 5x_3 = 0$

$-6x_3 = 0$

$x_3 = 0$

$-x_1 + 2 = 0$

$x_1 = 2$

List the eigenvalues of

$$A = \begin{pmatrix} -8 & -3 \\ 30 & 11 \end{pmatrix}$$

in ascending order.

$\lambda_1 =$ 1

$\lambda_2 =$ 2

Complete the coordinates of x to obtain an eigenvector of A associated with λ_1. (Give the result in decimal, rounded to 2 decimal places.)

$\lambda_1 = 1$

$x_1 = 1$

$x_2 =$ −3

$$\begin{vmatrix} -8-\lambda & -3 \\ 30 & 11-\lambda \end{vmatrix} = 0$$

$(-8-\lambda)(11-\lambda) - (-90) = 0$

$-88 + 8\lambda - 11\lambda + \lambda^2 + 90 = 0$

$\lambda^2 - 3\lambda + 2 = 0$

$(\lambda - 2)(\lambda - 1) = 0$

$\lambda = 1, 2$

$$\begin{pmatrix} -9 & -3 \\ 30 & 10 \end{pmatrix} \begin{pmatrix} 1 \\ x_2 \end{pmatrix} = 0$$

$-9 - 3x_2 = 0$

$30 + 10x_2 = 0$

$3x_2 = -9$

$x_2 = -3$

Let

$$A = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & -5 & -5 \\ 0 & 4 & 5 \\ 0 & 0 & -3 \end{pmatrix}.$$

Give the eigenvalues of A in ascending order.

-3

-2

1

Give the eigenvalues of B in ascending order.

-3

-1

4

$\det(A) = (-2-\lambda)(1-\lambda)(-3-\lambda) = 0$

$-2-\lambda = 0$ $\quad 1-\lambda = 0$ $\quad -3-\lambda = 0$

$\lambda = -2$ $\quad \lambda = 1$ $\quad \lambda = -3$

$\det(B) = (-1-\lambda)(4-\lambda)(-3-\lambda) = 0$

$-1-\lambda = 0$ $\quad 4-\lambda = 0$ $\quad -3-\lambda = 0$

$\lambda = -1$ $\quad \lambda = 4$ $\quad \lambda = -3$

If

$$x = \begin{pmatrix} -10 \\ 0 \\ 1 \end{pmatrix}, \quad \text{and} \quad y = \begin{pmatrix} -5 \\ 2 \\ -1 \end{pmatrix}$$

then the dot product of x and y is

Answer: 49

$\langle x, y \rangle = 0$

$(-10)(-5) + (1)(-1) = 50 - 1 = 49$

Calculate the Euclidean norm of

$$x = \begin{pmatrix} -5 \\ 7 \\ -2 \end{pmatrix},$$

rounded to 2 decimal places.

Answer: 8.83

$\sqrt{25 + 49 + 4} = \sqrt{78} = 8.83$

List the eigenvalues of

$$A = \begin{pmatrix} 18 & -8 \\ 40 & -18 \end{pmatrix}$$

in ascending order.

-2

2

$\begin{vmatrix} 18-\lambda & -8 \\ 40 & -18-\lambda \end{vmatrix}$

$(18-\lambda)(-18-\lambda) - (-320) = 0$

$-324 - 18\lambda + 18\lambda + \lambda^2 + 320 = 0$

$\lambda^2 - 4 = 0$

$(\lambda+2)(\lambda-2) = 0$

$\lambda = \pm 2$

Calculate the Euclidean norm of

$$x = \begin{pmatrix} 10 \\ 1 \\ -1 \end{pmatrix},$$

rounded to 2 decimal places.

Answer: 10.1

$\sqrt{10^2+1^2+(-1)^2} = \sqrt{102} = 10.099\cdots = 10.1$

List the eigenvalues of

$$A = \begin{pmatrix} -7 & 3 \\ -30 & 12 \end{pmatrix}$$

in ascending order.

small → large

$\lambda_1 =$ 2

$\lambda_2 =$ 3

Complete the coordinates of x to obtain an eigenvector of A associated with λ_1. (Give the result in decimal, rounded to 2 decimal places.)

$x_1 = 1$ $\lambda_1 = 2$

$x_2 =$ 3

$A = \begin{pmatrix} -7 & 3 \\ -30 & 12 \end{pmatrix}$

$\begin{vmatrix} -7-\lambda & 3 \\ -30 & 12-\lambda \end{vmatrix} = 0$

$(-7-\lambda)(12-\lambda) - (-90) = 0$

$-84 + 7\lambda - 12\lambda + \lambda^2 + 90 = 0$

$\lambda^2 - 5\lambda + 6 = 0$

$(\lambda-2)(\lambda-3) = 0$

$\lambda = 2, 3$

$\lambda_1 = 2$

$\begin{pmatrix} -7-2 & 3 \\ -30 & 12-2 \end{pmatrix} \to \begin{pmatrix} -9 & 3 \\ -30 & 10 \end{pmatrix}\begin{pmatrix} 1 \\ x_2 \end{pmatrix} = 0$

$\begin{cases} -9 + 3x_2 = 0 \\ -30 + 10x_2 = 0 \end{cases}$

$3x_2 = 9$

$x_2 = 3$

$\lambda = 1$ is an eigenvalue of

$$A = \begin{pmatrix} -3 & 4 & 4 \\ -1 & 1 & -1 \\ -1 & 2 & 4 \end{pmatrix}.$$

Complete the vector x to get an eigenvector corresponding to λ. (Round the result to 2 decimal places.)

$x_1 =$ -1

$x_2 =$ -2

$x_3 = 1$

$\lambda = 1$

$\begin{pmatrix} -3-1 & 4 & 4 \\ -1 & 1-1 & -1 \\ -1 & 2 & 4-1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ 1 \end{pmatrix} = 0$

$\begin{cases} \cancel{-4x_1 + 4x_2 + 4 = 0} \\ -x_1 + 0 - 1 = 0 \\ -x_1 + 2x_2 + 3 = 0 \end{cases}$

$x_1 = -1$

$-(-1) + 2x_2 + 3 = 0$

$2x_2 = -4$

$x_2 = -2$

Let

$$A = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 4 & 1 \\ 0 & -3 & -5 \\ 0 & 0 & 3 \end{pmatrix}.$$

Give the eigenvalues of A in ascending order.

-2

-2

4

Give the eigenvalues of B in ascending order.

-3

-1

3

$\det(A) = (4-\lambda)(-2-\lambda)(-2-\lambda) = 0$

$4-\lambda = 0$, $\lambda = 4$; $-2-\lambda = 0$, $\lambda = -2$; $-2-\lambda = 0$, $\lambda = -2$

$\det(B) = (-1-\lambda)(-3-\lambda)(3-\lambda) = 0$

$-1-\lambda = 0$, $\lambda = -1$; $-3-\lambda = 0$, $\lambda = -3$; $3-\lambda = 0$, $\lambda = 3$

If

$$x = \begin{pmatrix} -1 \\ -6 \\ -5 \end{pmatrix}, \quad \text{and} \quad y = \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$$

then the dot product of x and y is

Answer: 3

$(-1)(-1) + (-6)(-2) + (-5)(2) = 1 + 12 - 10 = 3$

Find the value of α, if the vectors x and y are orthogonal.

$$x = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad y = \begin{pmatrix} 12 \\ \alpha \end{pmatrix}$$

Answer: -4

$\langle x, y \rangle = 0$

$12 + 3a = 0$

$3a = -12$

$a = -4$

List the eigenvalues of

$$A = \begin{pmatrix} 9 & -4 \\ 20 & -9 \end{pmatrix}$$

in ascending order.

-1

1

$\begin{vmatrix} 9-\lambda & -4 \\ 20 & -9-\lambda \end{vmatrix}$

$(9-\lambda)(-9-\lambda) - (-80) = 0$

$-81 - 9\lambda + 9\lambda + \lambda^2 + 80 = 0$

$\lambda^2 - 1 = 0$

$(\lambda+1)(\lambda-1) = 0$

$\lambda = -1, 1$

Let

$$x = \begin{pmatrix} -2 \\ 8 \\ 4 \end{pmatrix}, \quad y = \begin{pmatrix} 0 \\ -8 \\ \alpha \end{pmatrix}, \quad z = \begin{pmatrix} \beta \\ \gamma \\ 8 \end{pmatrix}$$

Find the values of α, β, γ to be x, y, z pairwise orthogonal.

$\alpha =$ 16

$\beta =$ 80

$\gamma =$ 16

$\langle x, y \rangle = 0$: $-64 + 4a = 0 \longrightarrow 4a = 64 \rightarrow a = 16$

$\langle x, z \rangle = 0$: $-2\beta + 8\gamma + 32 = 0 \rightarrow -2\beta + 128 + 32 = 0 \rightarrow 2\beta = 160 \rightarrow \beta = 80$

$\langle z, y \rangle = 0$: $-8\gamma + 8a = 0 \rightarrow -8\gamma + 128 = 0 \rightarrow 8\gamma = 128 \rightarrow \gamma = 16$